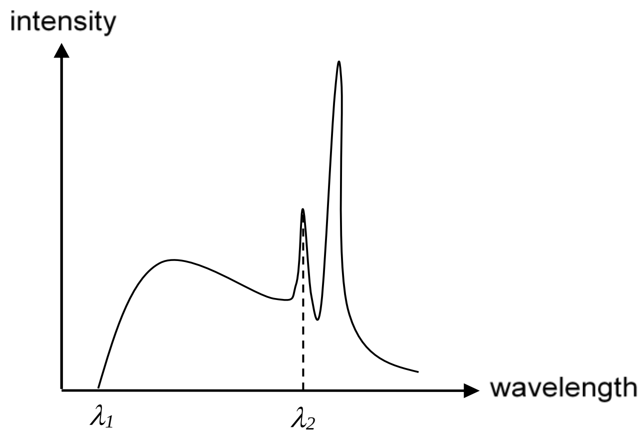
28 The graph below shows the variation of X-ray intensity with wavelength emitted from an X-ray tube.



The target metal and the accelerating voltage are varied. Which wavelengths will be affected by the changes?

	Changing the target metal	Changing the accelerating voltage
A	λ_1	λ_1
B	λ_1	λ_2
C	λ_2	λ_2
D	λ_2	λ_1

Answer: D

The minimum wavelength, λ_1, is affected by the potential difference which accelerates the electron and causes it to gain KE. The electron loses all its KE in the form of an X ray photon with minimum wavelength through a single head-on collision with the target atom.

λ_2 is located at one of the peaks. Changing the target metal will affect the characteristic wavelength of the peaks which is *dependent* on the target material. Since each element has a unique set of electron energy levels, the energy differences between these levels are also characteristic of the material.

29	One nuclear reaction that can take place in a nuclear reactor may be represented by the